

MIHYE KANG

Technical Program Management · Cloud Migration & AI Platform Delivery

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SUMMARY

17 years in enterprise IT at KT Group, 11 of them leading technical programs. Works where no delivery standard exists yet: builds the operating model, structures technical options until the forum reaches a decision, and moves organizations that do not report to her to a delivered result. Recent scope includes a 12-month, ~90-service Azure migration (on time, zero defects) and a three-party AI media agent integration program currently in delivery. CKA and AWS Solutions Architect Associate certified; authors Kubernetes manifests directly.

PROGRAM DELIVERY HIGHLIGHTS

Dimension	Evidence
Ambiguity to structure	Owned the 6 services the corporate Azure migration guide did not cover; built the discovery baseline, the manifest standard, and the progress metric from nothing
Delivery outcome	Azure 2025: 0 defects, on-time milestones, no added headcount; 99.8% manifest completion tracked to closure
Technical complexity to decision	Ran the architecture forum across KT, KTC, and Microsoft; structured 5 candidate options to a single selected non-HTTP (TCP/MQTT) architecture, adopted as the project-wide standard
Organizations to execution	Delivered through parties with no reporting line: 17 consortium organizations (2019) and a three-party KT / KTDS / vendor program (2026); 3 concurrent service lines (2022-2023)
AI in delivery	STT and LLM cascade pipeline design; AI-assisted log analysis for root-cause narrowing; Agentic AI certification program completed 2026.06

PROFESSIONAL EXPERIENCE

KTDS (KT Group IT subsidiary) · Program Manager / Project Lead · 2015.01 – Present

KT Azure Mass Migration · Non-ITO Project Lead · 2025.01 – 2025.12

Context. Company-wide migration of approximately 90 services to Azure. Owned the 6 non-standard Non-ITO services, for which the corporate migration guide provided no applicable criteria.

- **Requirements baseline.** Designed a 16-item discovery questionnaire (current architecture, load profile, verification criteria, operating model, data sizing, security approval path) and applied it uniformly across all 6 services to produce comparable inputs. Classified absent test scenarios as a missing success criterion rather than a documentation gap and required authoring before kickoff. Pre-cleared firewall requirements, eliminating migration-day wait states.
- **Architecture decision.** The corporate guide assumed HTTP-only traffic, leaving no decision basis for a persistent TCP/MQTT integration. Led the technical forum across the architecture team, KT, KTC, and Microsoft, evaluated 5 candidate architectures through a four-stage filter (technical feasibility, operational risk, change scope, environment fit), and selected AKS internal load balancer with pod-level TLS over public-preview App Gateway TCP/TLS and options requiring client or firmware rework. Adopted as the project-wide standard.
- **Delivery control.** Identified the absence of any means to track per-pod manifest progress as the problem itself. Designed a Confluence progress tracker operated as a management metric rather than a record, and tracked manifest completion to 99.8%.
- **Critical-path intervention.** When manifest authoring became the primary delay driver, learned the Kustomize standard and authored manifests directly rather than waiting for headcount. Obtained CKA during the program. Converted the architecture team's principles into a manifest standard partner vendors could apply without interpretation.
- **Incident analysis.** Narrowed Rancher logs to the failure window and used AI-assisted analysis to reduce the hypothesis space on a StatefulSet startup failure; verified root cause as etcd state not reflecting a forced VM shutdown.

Results. All migration milestones delivered on time with no additional resources. Zero defects. All migrated services operating in production. TCP integration architecture adopted as project-wide standard.

Stack. Azure, AKS, Rancher, Kustomize, ArgoCD, Microsoft Entra ID, TCP/MQTT, MariaDB to MySQL, Jira, Confluence

KT ADD (AI Media Agent Integration) · Program Manager · 2026.05 – Present

Context. Three-party program: KT as product owner, KTDS engineering, and a partner vendor. No formal command authority over the other parties.

- **Integration contract.** Surveyed the full field set of external sources including TMDB, excluded low-signal fields, and finalized the media agent integration specification around `contentId`, `title`, `releaseDate`, and `synopsis`.

- **Dependency reduction by design.** Adopted a structure where KTDS loads files and the media agent retrieves them by API, and assigned data quality judgment and display policy to the agent side. Neither party is bound to the other's release schedule. Cross-organization dependency is reduced more durably by architecture than by coordination cadence.
- **Constraint separation.** With no GPU allocation, self-hosted open-source inference (QWEN) was not viable; converted to API-based Gemini Flash single-path processing. Separated the cause, business-unit budget rather than a technical limit, and returned it as a documented product-owner decision item with named ownership and a response deadline. Scope, schedule, and resource trade-offs were returned to KT on the same basis.
- **Program metrics.** Designed and tracks weekly: blocker age, decision lead time, and deadline-specification rate. Status narrative does not surface risk; elapsed days do.
- **Forum operation.** Ran a dual-track model for a client that resists procedural communication: conversational externally, strict decision and risk registers internally. Post-meeting summary emails serve as the decision log where Confluence is unavailable. Introduced a two-stage response protocol (immediate provisional answer, then dated confirmation) to eliminate no-response intervals.

Results. Media agent integration specification finalized. Contents Agent infrastructure design complete, environment build in progress. Zero blockers aged beyond 7 days.

Stack. Whisper (self-hosted STT), Gemini Flash API, LLM orchestration, TMDB external API, Confluence

| KTDS Company-wide ITO Business Structure Innovation Task Force · 2024.06 – 2024.12

Owned collection and analysis of field-level ITO contract problems. The difficulty was recognition, not collection: scope growth without corresponding compensation had settled into accepted practice, so survey instruments returned nothing. Contacted team leads directly through referral, then collected, classified, and pattern-analyzed three years of contract cases until individual complaints resolved into recurring types addressable at company level. Supplied field interview data into a KOSA rate-based simulation run with external consultants and authored the closing report.

Results. Workload measurement established as a process that kept executing for approximately 6 months after roll-off. Fed into the 2026 ITO rate increase.

| KT Telecom Big Data Platform Portal · Portal Project Lead · 2019.07 – 2020.02

Context. 17 external consortium organizations, none under reporting or command authority, required to produce conforming metadata deliverables.

- Translated DCAT, an unfamiliar international metadata standard, into directly applicable form: separated 21 fields into mandatory and optional, specified authoring rules, and distributed guide, template, examples, naming conventions, and deadlines as a single package. Operated asynchronously against the common guide, with direct contact for non-responding organizations.
- When the standard changed mid-program from RDF to xlsx, defined an interim integration format rather than waiting for the national agency's final ruling, keeping partner work uninterrupted.

Results. 140 metadata records collected. Analysis, visualization, and ML linkage process designed. Launched within the ministry demonstration window with no execution gap.

Failure and the standard that came from it. A display-resolution mismatch in design deliverables caused rework: receipt of design images had been treated as completion, with no defined acceptance criterion. The correction was procedural, not personal. UI sizing verification was applied at Senior Care (2021) and Home Portal (2023), and test scenarios became a mandatory item in the Azure migration discovery questionnaire (2025).

| KT AI First Contact Center (AICC) · Scenario Development Project Lead · 2020.03 – 2020.12

- The business side reframed the chatbot from inquiry handling to transaction completion. Led integration design and implementation for billing and collections core-system linkage, including the failure handling and consistency conditions monetary transactions require, delivering overdue-balance check through payment within a single conversation.

Results. KRW 32M collected within 3 weeks of launch.

| Additional programs

Period	Program	Role and outcome
2022.03 – 2023.02	KT GiGA Genie backend platform	PM. Ran 3 service lines concurrently (backend, personalization, senior care) to on-schedule completion; assumed commercial launch directly after a colleague's departure. Proposed MicroStrategy adoption and ran vendor selection and contracting.
2023.10 – 2024.12	KT Senior Care web application	PM. Led web application development, delivered within GiGA Genie TV.
2021.01 – 2022.02	KT Senior Care	PM. New service build and transition to managed operations.
2018.04 – 2019.06	KT IoT Support	SM PM. Led operations transition. Improved dashboard query performance from 200s to 1s (company incentive award). Restructured NB-IoT monitoring screens, reducing CPU and memory utilization.

R&B Software · Developer and Operations · 2009.01 – 2014.12

Siebel CRM operations and development for KT sales systems. Built approximately 50 to 60 order-to-legacy integration interfaces under Agile sprint

delivery. Module lead for the B2C CRM marketing module during the next-generation sales system program (Siebel to Nexacro), delivering soft launch and production launch; received CEO commendation.

AI IN PROGRAM MANAGEMENT

- **Delivery workflow.** AI-assisted log analysis to narrow root-cause hypotheses and reduce investigation time (Azure StatefulSet failure, 2025).
- **Architecture judgment.** Finalized the Whisper STT to Contents Agent orchestration to Gemini Flash cascade pipeline with the engineering team, deciding the self-hosted versus API trade-off under a hard resource constraint.
- **Formal training.** Agentic AI program completed 2026.06 (LangChain, LangGraph, RAG, MCP, multi-agent). CLI-based agent development environments configured on work and personal machines, including corporate proxy and TLS inspection resolution.

TECHNICAL PROFILE

Area	Detail
Cloud and infrastructure	Azure, Kubernetes (AKS), Rancher, Kustomize, ArgoCD, Microsoft Entra ID, Azure Landing Zone, Application Gateway, Microsoft Defender for Cloud
Data	SQL (PostgreSQL, MySQL, MariaDB), query performance tuning, MicroStrategy (BI) adoption and vendor selection, metric design and weekly tracking
AI	Whisper (STT), Gemini API, LLM orchestration, Azure AI Foundry / Azure OpenAI, AI-assisted log analysis
Program management	Structured discovery interviews, integration specification ownership, architecture decision forums, risk and decision registers, NFR logging, milestone and dependency management
Tooling	Jira, Confluence
Development background	Java/JSP, Siebel CRM, Nexacro

CERTIFICATIONS AND EDUCATION

Item	Detail
Certifications	Certified Kubernetes Administrator, CKA (2026.01) · AWS Solutions Architect Associate (2024.05) · Siebel 8 Consultant Expert (2011) · Engineer Information Processing, KR national certification (2009)
Education	B.S. Computer Science, Hanshin University (2009.02)

LANGUAGES

- **Korean.** Native.
- **English.** Currently studying; working toward professional proficiency in written and spoken English.